A Moser diffeomorphism pulling omega back to omega_0, fixing the boundary

The Univalent Foundations Program

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Editorial note. The counted statement is the author's labelled lemma 'l:pullback_to_nu0' (source lines 1501-1504); the proof environment starts after one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's displayed collar equation for omega_0, invoked inside the retained proof; it is an equation, so it consumes no theorem number in the retained sequence.

Editorial note. The retained counter values (section 3, theorem offset 6) make the counted lemma print as Lemma 3.7; this article carries no label-number dump, so the number, and the numbers of the two navigation pointers, were inferred by counting the theorem-class environments of section 3 and are reported with that caveat.

Editorial note. The two external references are navigation pointers to the author's transversality assumption and to the normalisation lemma cited for its first two parts; neither is load-bearing in place of the counted proof, which is retained in full.

Editorial note. The retained preamble is a minimal editorial adapter supplying only the plain text abbreviations that the publisher class used to define, followed by the native author macro block of the pinned source.

Editorial scope. One original result of the pinned publisher article, counted once: the Moser diffeomorphism pulling omega back to omega_0 and fixing the boundary (author label l:pullback_to_nu0).

Allowed standard prerequisites (editorial).

- C2-closeness of contact forms. lambda C2-close to lambda_0 implies nu and omega close in the corresponding norms.. It is the hypothesis and the first remark of the proof at source line 1507.
- Moser's trick with boundary conditions. One looks for an isotopy fixing the boundary whose pull-backs equal omega_0, generated by a vector field solving X_t interior omega_t = nu_0 - nu + d f_t.. It is the framework of the whole proof, source lines 1509-1518.
- Collar model on the annulus. On a collar A_C the forms can be written with the functions h_1, h_2, h_3 and the differential identity partial_r(r h_2) - partial_s h_1 = r(h_3 - 1).. They are used to choose f and to estimate R_t, source lines 1525-1577.

Source reference. Original definition 3.4: sources/jep-2022-195-contact-systolic/pinned-original-native.tex, source lines 1405–1405. Original result invoked by the retained proof. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number of this article's own results was inferred by counting the theorem-class environments of its section of the pinned source, which carries no label-number dump; it is not confirmed by the retained bibliography.

Source reference. Original lemma 3.3: sources/jep-2022-195-contact-systolic/pinned-original-native.tex, source lines 1289–1289. Original result invoked by the retained proof. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number of this article's own results was inferred by counting the theorem-class environments of its section of the pinned source, which carries no label-number dump; it is not confirmed by the retained bibliography.

$$\omega_0|_{A_C} = \underbrace{\frac{4\pi}{\alpha_1}\big(q_0 + (\alpha_1'/\alpha_1)p_0\big)}_{(*)}\, r\, dr \wedge ds. \tag{1}$$

Lemma 3.7. *If λ is C^2-close enough to λ_0, then there exists a diffeomorphism $\kappa : \Sigma \to \Sigma$ such that $\kappa|_{\partial\Sigma} = \mathrm{id}$ and $\kappa^*\omega = \omega_0$. Moreover, κ C^1-converges to the identity as λ C^2-converges to λ_0.*

Proof. Note that the smallness of $\|\lambda - \lambda_0\|_{C^2}$ implies the smallness of $\|\nu - \nu_0\|_{C^2}$ and $\|\omega - \omega_0\|_{C^1}$. Assumption 3.4 guarantees that ω is a symplectic form in the interior of Σ inducing the same orientation as ω_0. Therefore, the 2-forms $\omega_t := t\omega + (1-t)\omega_0$ are symplectic on $\mathrm{int}(\Sigma)$ for every $t \in [0,1]$. They are actually uniformly C^1-close to ω_0 when $\|\lambda - \lambda_0\|_{C^2}$ is small.

We look for an isotopy $\kappa_t : \Sigma \to \Sigma$ such that $\kappa_t|_{\partial\Sigma} \equiv \mathrm{id}$ and $\kappa_t^*\omega_t = \omega_0$. We build the time-dependent vector field X_t realizing such isotopy, i.e. $\frac{d}{dt}\kappa_t = X_t \circ \kappa_t$. By differentiating $\kappa_t^*\omega_t$ with respect to t, we obtain

$$0 = \tfrac{d}{dt}\kappa_t^*\omega_t = \kappa_t^*\big(d(X \lrcorner \omega_t) + \omega - \omega_0\big) = \kappa_t^* d\big(X \lrcorner \omega_t + \nu - \nu_0\big). \tag{2}$$

We define X_t on $\mathrm{int}(\Sigma)$ by the equation

$$X_t \lrcorner \omega_t = \nu_0 - \nu + df_t \tag{3}$$

for a suitable C^2-small smooth function $f : \Sigma \times [0,1] \to \mathbb{R}$ to be determined. A suitable choice of f will guarantee that X_t has a smooth extension to the whole Σ with $X_t|_{\partial\Sigma} \equiv 0$.

For every connected component C of $\partial\Sigma$, we fix a collar neighborhood $A_C \subset \Sigma$ so that, with the usual suitable identification $[0,\rho) \times S^1$, the differential forms ω_0 can be written as in (1). Actually, up to rescaling the interval $[0,\rho)$, we can even write $\omega_0|_{A_C}$ as

$$\omega_0|_{A_C} = -r\, dr \wedge ds,$$

where $r \in [0,\rho)$ and $s \in S^1$. Lemma 3.3(i-ii) implies $\nu|_C = \nu_0|_C$ and $\omega_z = (\omega_0)_z$ for all $z \in C$. Therefore we can write

$$(\nu_0 - \nu)|_{A_C} = h_1\, dr + r\, h_2\, ds, \qquad \omega|_{A_C} = -r\, h_3\, dr \wedge ds,$$

for some smooth functions $h_i : A_C \to \mathbb{R}$. If ν_0 and ν are C^2-close, the function h_1 is C^2-small, while h_2 is C^1-small. Moreover, since ω_0 and ω are C^1-close, the function $1 - h_3(0, \cdot)$ is C^0-small. In particular, up to choosing the annulus A_C small enough, h_3 is strictly positive on the whole A_C. Since $d(\nu_0 - \nu) = \omega_0 - \omega$, we have

$$\partial_r(r\, h_2) - \partial_s h_1 = r(h_3 - 1). \tag{4}$$

The two-form $\omega_t|_{A_C}$ is given by

$$\omega_t|_{A_C} = -r\,(t(h_3 - 1) + 1)\, dr \wedge ds,$$

and if we write the vector field $X_t|_{A_C}$ in (r, s) coordinates as $X_t = R_t\, \partial_r + S_t\, \partial_s$, Equation (3) becomes

$$R_t = -\frac{r\, h_2 + \partial_s f}{r\,(t(h_3 - 1) + 1)}, \qquad S_t = \frac{h_1 + \partial_r f}{r\,(t(h_3 - 1) + 1)}.$$

We now choose $f : \Sigma \to \mathbb{R}$ to be a smooth function such that $f|_{\partial\Sigma} \equiv 0$ and, on any collar neighborhood $A_C = [0, \rho) \times S^1$ as above, satisfies

$$f(r, s) = \begin{cases} -\displaystyle\int_0^r h_1(x, s)\, dx, & \text{if } r \le \tfrac{1}{3}\rho, \\ 0, & \text{if } r \ge \tfrac{2}{3}\rho. \end{cases}$$

We shall choose such an f so that $\|f\|_{C^2} \le \text{const}\, \|h_1\|_{C^2}$, and in particular f is C^2-small since ν and ν_0 are C^2-close. With this choice of f, we have $S_t(r, s) = 0$ if $r \le \frac{1}{3}\rho$. As for the function R_t, for all $r \le \frac{1}{3}\rho$ Equation (4) implies

$$\begin{aligned} R_t(r, s) &= -\frac{r\, h_2(r, s) + \partial_s f(r, s)}{r\,(t(h_3 - 1) + 1)} = -\frac{r\, h_2(r, s) - \int_0^r \partial_s h_1(x, s)\, dx}{r\,(t(h_3 - 1) + 1)} \\ &= -\frac{r\, h_2(r, s) + \int_0^r \big(x(h_3(x, s) - 1) - \partial_x(x\, h_2(x, s))\big)\, dx}{r\,(t(h_3 - 1) + 1)} \\ &= -\frac{\frac{1}{r}\int_0^r x(h_3(x, s) - 1)\, dx}{t(h_3 - 1) + 1}. \end{aligned}$$

We already know that the function $t(h_3 - 1) + 1$ appearing in the denominator of the above equation is nowhere vanishing. As for the numerator, we can rewrite the integral as

$$\int_0^r x(h_3(x, s) - 1)\, dx = r^2\, h_4(r, s)$$

for some C^0-small smooth function $h_4 : A_C \to \mathbb{R}$ such that $h_4(0, s) = h_3(0, s) - 1$. Therefore

$$R_t(r, s) = -\frac{r\, h_4(r, s)}{t\,(h_3 - 1) + 1}.$$

From this expression we readily infer that R_t is C^1-small, extends smoothly to the whole Σ, and $R_t|_{\partial\Sigma} \equiv 0$. Summing up, we obtained a C^1-small smooth vector field X_t on Σ satisfying (3) and $X|_{\partial\Sigma} \equiv 0$. Its flow κ_t is C^1-small for all $t \in [0, 1]$, and satisfies $\kappa_t|_{\partial\Sigma} = \text{id}$ and, by (2), $\kappa_t^* \omega_t = \omega_0$. □